

Dr. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Winter Examinations 2022

Course: B. Tech. Branch : Computer Science & Engineering Semester : III

Subject Code & Name: BTCOC302 _ Discrete Mathematics

Max Marks: 60

Date: 11/03/2023

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly

	(BTLevel)	Marks
Q.1 Solve Any Two of the following.		12
A) Among the integers 1 to 1000:	Apply	6
1) How many of them are not divisible by 3, or by 5, or by 7?		
2) How many are not divisible by 5 and 7 but divisible by 3.		
B) Prove that $1 + 2 + 2^2 + \dots + 2^n = 2^{n+1} - 1$, where n is a nonnegative integer by using mathematical induction	Understand	6
C) Let $S(x)$ be the predicate "x is a student," $F(x)$ the predicate "x is a faculty member," and $A(x, y)$ the predicate "x has asked y a question," where the domain consists of all people associated with your school. Use quantifiers to express each of these statements.	Understand	6
I. Lois has asked Professor Michaels a question.		
II. Every student has asked Professor Gross a question.		
III. Every faculty member has either asked Professor Miller a question or been asked a question by Professor Miller.		
IV. Some student has not asked any faculty member a question.		
V. There is a faculty member who has never been asked a question by a student		
VI. There is a faculty member who has asked every other faculty member a question		
Q.2 Solve Any Two of the following.		12
A) Let $A = \{p, q, r\}$. Show $(P(A), \text{subset})$ is a poset and draw its Hasse diagram.	Apply	6
B) How many permutations of the letters a, b, c, d, e, f, g contain neither the pattern bge nor eaf?	Apply	6
C) Determine whether each of these functions is one-to-one, onto, both or not a function. Justify your answer	Apply	6

1) from $\{a, b, c, d\}$ to itself

i. $f(a) = b, f(b) = a, f(c) = c, f(d) = d$

ii. $f(a) = b, f(b) = b, f(c) = d, f(d) = c$

2) from $\mathbb{Z}$ to $\mathbb{Z}$

i. $f(n) = n + 1$

ii. $f(n) = n^3$

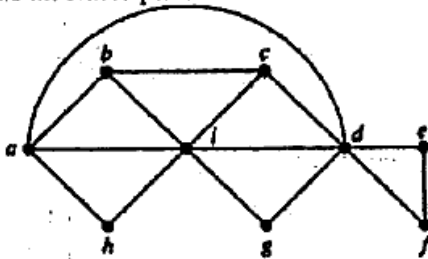
12

Q.3 Solve Any two of the following.

Apply

6

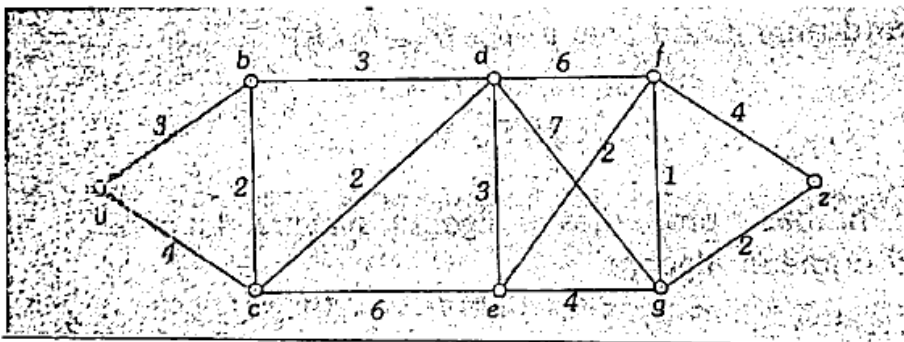
- A) Determine whether the given graph has an Euler circuit. Construct such a circuit when one exists. If no Euler circuit exists, determine whether the graph has an Euler path and construct such a path if one exists.



- B) Find shortest path between a to z by Dijkstra's Method.

Understand

6

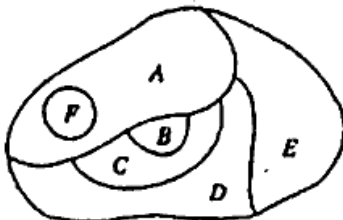


- C) For the following maps <https://www.pyqonline.com>

Apply

6

- 1) Draw the planar graph
- 2) Color the vertices of graph.
- 3) Find the chromatic number



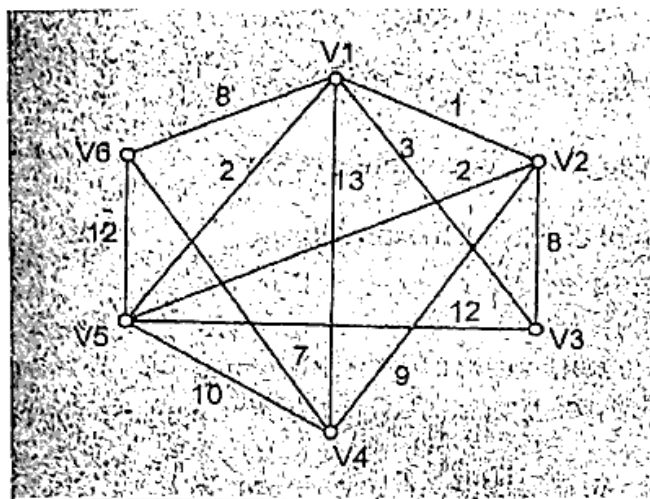
Q.4 Solve Any Two of the following.

- A) Find Minimum spanning tree for following graph by using Kruskal's algorithm.

Apply

12

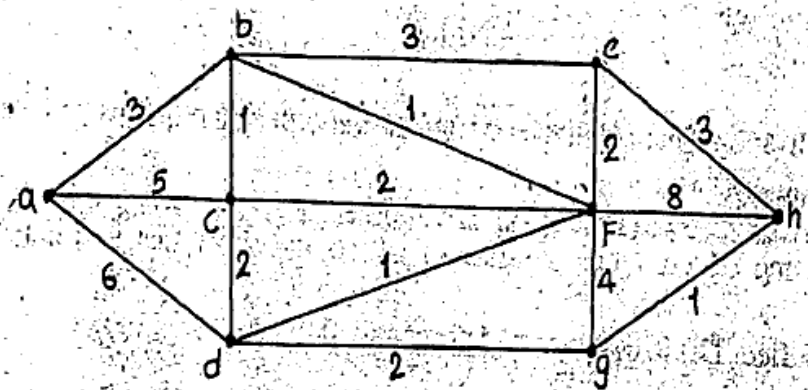
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B) Find Minimum spanning tree for following graph by Prim's Algorithm.

Apply

6



C) Define Rooted tree, Balanced Tree and Binary search tree. Explain with example

Understand

6

Q. 5 Solve Any two of the following.

12

A) Define Semigroup, Monoid and Group. Explain with example

Remember

6

B) Let $(G, \oplus)$ is an algebraic system where $G = \{1101, 0000, 1001, 0100\}$ prove that $(G, \oplus)$ is a group.

Apply

6

C) Define Ring. Prove that An algebraic system $(Z, +, \cdot)$ is a ring, where $+$ is addition and $\cdot$ means multiplication operation. i.e. $a + b$ as the sum of a and b in Z , and $a \cdot b$ as the product of a and b in Z .

Apply

6

*** End ***